

Calculus of Variations Notes

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Figure 1: Variations - Needle pierces the mystery. < |

1 Introduction to variations

In this section, we first introduce some basic concepts which are indispensable for our concerned subjects and related problems. And then giving a typical problem example to help readers understand preliminarily what kind of problem are we studying. So be relaxed, and enjoy this section:

1.1 Basics summary

The objects what we mainly studies throught out the book:

Definition 1. Given a set M of functions, the map $I : M \rightarrow \mathbb{C}$ is called a **functional**.

Some example to tell if it a map is a functional:

Example. $I_1(u) = \max_{x \in \Omega} |u(x)|$, $I_2(u) = u(x_0)$, $I_3(u) = \int_{\Omega} [|\nabla u(x)|^2 - F(u(x))] dx$ are all functional.

Example. $\forall M$ and $f : \mathbb{R} \rightarrow \mathbb{R}$, $I_4(u) = f(u(x))$ is not functional.

A typical example problem to introduce the subject what we studies:

Problem 1. (Brachistochrone) Given two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$ in a vertical plane, where $x_1 < x_2$, $y_1 > y_2$. A particle slides along a smooth curve connecting these two points under gravity alone. Assuming initial velocity is zero, ask along which curve the sliding time is the shortest

Solution. Let $u \in C^1[x_1, x_2]$, $(x, u(x)) \mid x \in [x_1, x_2], u(x_i) = y_i, i = 1, 2$ is a curve connecting A, B . Since we have $\begin{cases} \frac{1}{2}mv^2 = mgh, \\ v = \frac{ds}{dt}, \end{cases}$ so $v = \sqrt{2g(y_1 - u(x))}$, and $dt = \frac{ds}{v} = \sqrt{\frac{1+|u'(x)|^2}{2g(y_1 - u(x))}} dx$, total time $T = \int_{x_0}^{x_1} dt = \frac{1}{\sqrt{2g}} \int_{x_0}^{x_1} \sqrt{\frac{1+|u'(x)|^2}{y_1 - u(x)}} dx$.

Let $M = \{u \in C^1([x_1, x_2]) \mid u(x_i) = y_i, i = 1, 2\}$, then the mapping $M \rightarrow \mathbb{R}$, $u \mapsto T$ is a "functional". Here u is the independent variable, $T = T(u)$ is the dependent variable. We want to find u in M such that T is minimized.

2 Euler-Lagrange Equation

2.1 Summary and comprehension

Similar with the derivation process of the necessary conditions for the extreme value of function, we could get almost same progress for functional. And since the variant of target functional now is function, so we should introduce a Lemma (du Bois-Reymond) to deal with it, and after that the necessity of extreme value of functional is appeared.

Lemma 2.1. (*du Bois-Reymond*)

If $\psi \in C[t_0, t_1]$, and

$$\int_J \psi(t) \cdot \lambda'(t) dt = 0, \quad \forall \lambda \in C_0^1(J), \quad J = [t_0, t_1],$$

$C_0^1(J) = C_0^1(J, R^1) = \{u \in C^1(J) : u(t_0) = u(t_1) = 0\}$, then $\psi = \text{const}$

Proof. [/Tech toolkit/](#) (Wonderful substitution!)

Let $c = \frac{1}{|J|} \int_J \psi(t) dt$, $\lambda(t) = \int_{t_0}^t (\psi(s) - c) ds$.

Then consider $\int_J (\psi(t) - c)^2 dt = \int_J \psi(t)(\psi(t) - c) dt = \int_J \psi(t) \lambda'(t) dt = 0$, thus $\psi(t) \equiv c$.